Exam : End Term Quiz
Subject : Maths2
Total Marks : 50.00
QP : 2024 Sep01: IIT M FN EXAM QDF1

Exam Mode | Learning Mode

View Question Paper Summary

QUESTION MENU

1 2 3 4 5 6 7 8 9
10 11 12 13 14 15 16 17 18
19 20 21 22 23 24 25 26 27
28

TIMER

00:28

CONTROLS

SUBMIT EXAM

Your Score

0.00 / 50.00

(0%)

Question 1 : 640653902368

Total Mark : 0.00 | Type : MCQ

THIS IS QUESTION PAPER FOR THE SUBJECT "FOUNDATION LEVEL : SEMESTER II: MATHEMATICS FOR DATA SCIENCE II (COMPUTER BASED EXAM)" ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT? CROSS CHECK YOUR HALL TICKET

TO CONFIRM THE SUBJECTS TO BE WRITTEN. (IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

OPTIONS :

- YES
- NO

Your score : 0

Discussions (0)

Question 2 : 640653902369

View Solutions (0)

Total Mark : 3.00 | Type : MCQ

Consider the two statements given below:

Statement-P: There exists a square matrix of order 2 such that $A^2 = 0$, $A^T = A$ and $A \neq 0$.

Statement-Q: There exists a square matrix of order 2 such that $A^2 = 0$ and $A \neq 0$.

OPTIONS :

- Statement-P is true, but statement-Q is false.
- Statement-P is false, but statement-Q is true.
- Both statements are true.
- Both statements are false.

Your score : 0

Discussions (0)

Question 3 : 640653902370

View Solutions (0)

Total Mark : 3.00 | Type : MCQ

Consider the following system of equations:

$$\begin{aligned} 2x_1 - x_2 + x_3 &= b_1 \\ x_1 + x_2 - 2x_3 &= b_2 \\ -3x_2 + 5x_3 &= b_3 \end{aligned}$$

Under which of the following conditions is the system consistent?

OPTIONS :

- $b_2 = 0$
- $b_1 = 2b_2$
- $b_3 = b_1 - 2b_2$
- $b_1 + b_2 + b_3 = 0$

Your score : 0

Discussions (0)

Question 4 : 640653902371

View Solutions (0)

Total Mark : 3.00 | Type : MCQ

Which of the following is an orthonormal basis for span$\{(1, 1, -1), (1, 2, 0)\}$?

OPTIONS :

- $\left\{\frac{1}{\sqrt{3}}(1, 1, -1), \frac{1}{\sqrt{2}}(0, 1, 1)\right\}$
- $\left\{\frac{1}{\sqrt{3}}(1, 1, -1), \frac{1}{\sqrt{5}}(1, 2, 0)\right\}$
- $\left\{\frac{1}{\sqrt{2}}(1, 0, 1), \frac{1}{\sqrt{2}}(-1, 0, 1)\right\}$
- $\left\{\frac{1}{\sqrt{2}}(0, 1, 1), \frac{1}{\sqrt{2}}(0, -1, 1)\right\}$

Your score : 0

Discussions (0)

Question 5 : 640653902372

View Solutions (0)

Total Mark : 3.00 | Type : MSQ

Let $S = \{u_1, u_2, u_3, u_4, u_5\}$ be a linearly independent subset of a vector space V. Select all true statements.

OPTIONS :

- [ ] There exists a basis B for V such that $S \subseteq B$
- [ ] $\dim(V) \geqslant 5$
- [ ] $\text{span}(S) = V$
- [ ] S is a subset of every basis B

Your score : 0

Discussions (0)

Question 6 : 640653902373

View Solutions (0)

Total Mark : 3.00 | Type : MSQ

Consider the function f:

$$f(x,y) = \begin{cases} \dfrac{y^4}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$$

If f_x, f_y denote the partial derivatives, select all true statements from the options given below.

OPTIONS :

- [ ] $f_x(0,0)$ does not exist.
- [ ] $f_y(0,0)$ does not exist.

- [ ] $f_x(0,0) = 0$
- [ ] $f_y(0,0) = 0$

Your score : 0

Discussions (0)

Question 7 : 640653902374

View Solutions (0)

Total Mark : 3.00 | Type : SA

Find the dimension of the affine subspace corresponding to the set of all solutions to the following system:

$$\begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Answer (Numeric):

Answer

Accepted Answer : 2

Your score : 0

Discussions (0)

Question 8 : 640653902375

View Solutions (0)

Total Mark : 3.00 | Type : SA

Let $v = (\sqrt{2}, -\sqrt{2}, 6) \in \mathbb{R}^3$. (a, b, c) is the vector obtained from v after rotating the XY-plane anti-clockwise by 45° about the Z-axis. Find the value of $a+b+c$. Enter the nearest integer as your answer.

Answer (Numeric):

Answer

Accepted Answer : 8

Your score : 0

Discussions (0)

Question 9 : 640653902376

View Solutions (0)

Total Mark : 3.00 | Type : SA

Evaluate $\lim\limits_{(x,y)\to(0,0)} \dfrac{\sin(4x^2+4y^2)}{x^2+y^2}$.

Answer (Numeric):

Answer

Accepted Answer : 4

Your score : 0

Discussions (0)

Question 10 : 640653902377

View Solutions (0)

Total Mark : 3.00 | Type : SA

Consider the following function:

$$f(x,y) = \begin{cases} \dfrac{x^2-xy}{\sqrt{x}-\sqrt{y}}, & x \neq y \text{ and } x \geqslant 0 \text{ and } y \geqslant 0 \\ c\ , & \text{otherwise} \end{cases}$$

For what value of c is the function f continuous at $(0,0)$?

Answer (Numeric):

Answer

Accepted Answer : 0

Your score : 0

Discussions (0)

Question 11 : 640653902378

View Solutions (0)

Total Mark : 3.00 | Type : SA

If the directional derivative of $f(x, y) = x^3y + xy^2 + 8$ at the point $(1, 2)$ in the direction $\left(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right)$ is $\frac{p}{\sqrt{2}}$, find the value of p.

Answer (Numeric):

Answer

Accepted Answer : 5

Your score : 0

Discussions (0)

Question 12 : 640653902379

View Solutions (0)

Total Mark : 3.00 | Type : SA

The equation of the tangent plane to the function $f(x, y) = e^x(x + y)$ at the point $(1, 0)$ is given by $Ax + By + C = z$. Find the value of $\frac{A+B+C}{e}$.

Answer (Numeric):

Answer

Accepted Answer : 2

Your score : 0

Discussions (0)

Question 13 : 640653902380

View Solutions (0)

Total Mark : 4.00 | Type : SA

If $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 5 \\ -4 & 9 & 5 \end{bmatrix}$, find the determinant of $A^3 - 3A^2 + 2A$.

Answer (Numeric):

Answer

Accepted Answer : 0

Your score : 0

Discussions (0)

Question 14 : 640653902392

Total Mark : 0.00 | Type : COMPREHENSION

Consider a linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$:

$$T(x, y, z) = (x - y + z, x + z, y - 2z)$$

Use the given standard basis for both domain and co-domain for matrix representation , Answer the sub-questions

Your score : 0

Question 15 : 640653902393

View Parent QN

View Solutions (0)

Total Mark : 1.00 | Type : MCQ

What is the matrix representation of T?

OPTIONS :

- $\begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & -2 \end{bmatrix}$
- $\begin{bmatrix} 1 & 1 & 0 \\ -1 & 0 & 1 \\ 1 & 1 & -2 \end{bmatrix}$

- $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
- $\begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

Your score : 0

Discussions (0)

Question 16 : 640653902394

View Parent QN | View Solutions (0)

Total Mark : 1.00 | Type : MSQ

Select all true statements.

OPTIONS :

- The kernel of T is $\{(0,0,0)\}$
- The kernel of T is $\mathbb{R}^3$
- The image of T is $\{(0,0,0)\}$
- The image of T is $\mathbb{R}^3$

Your score : 0

Discussions (0)

Question 17 : 640653902395

View Parent QN | View Solutions (0)

Total Mark : 2.00 | Type : MCQ

T^{-1} is a linear transformation that is the inverse of T. That is, if $T(x, y, z) = (a, b, c)$, then $T^{-1}(a, b, c) = (x, y, z)$.

Find the matrix representation of T^{-1},

OPTIONS :

- $\frac{1}{2}\begin{bmatrix} 1 & 1 & 1 \\ -2 & 2 & 0 \\ -1 & 1 & -1 \end{bmatrix}$
- $\frac{1}{2}\begin{bmatrix} 1 & -2 & -1 \\ 1 & 2 & 1 \\ 1 & 0 & -1 \end{bmatrix}$
- $\begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & -2 \end{bmatrix}$
- $\begin{bmatrix} -1 & -1 & 1 \\ -1 & 0 & 1 \\ 2 & 1 & -1 \end{bmatrix}$

Your score : 0

Discussions (0)

Question 18 : 640653902381

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Let $f : D \to \mathbb{R}$ be defined as follows:

$$f(x, y) = \sqrt{1 - x^2 - y^2}$$

where $D \subset \mathbb{R}^2$ is the domain of f.

Your score : 0

Question 19 : 640653902382

View Parent QN | View Solutions (0)

Total Mark : 1.00 | Type : MCQ

Choose the largest possible D that would be a valid domain from the options given below.

OPTIONS :

- $\mathbb{R}^2$
- $\{(x, y) : |x| \leqslant 1 \text{ and } |y| \leqslant 1;\ x, y \in \mathbb{R}\}$
- $\mathbb{R}^2 \setminus \{(x, y) : x^2 + y^2 < 1;\ x, y \in \mathbb{R}\}$
- $\mathbb{R}^2 \setminus \{(x, y) : x^2 + y^2 \leqslant 1;\ x, y \in \mathbb{R}\}$
- $\{(x, y) : x^2 + y^2 \leqslant 1;\ x, y \in \mathbb{R}\}$

Your score : 0

Discussions (0)

Question 20 : 640653902383

View Parent QN | View Solutions (0)

Total Mark : 2.00 | Type : MCQ

Which of the following is the range of f using the domain chosen from the previous question? Recall that $[a, b] = \{x : a \leqslant x \leqslant b;\ x \in \mathbb{R}\}$

OPTIONS :

- $[0, 1]$
- $[-1, 1]$

- $\mathbb{R}$
- $\mathbb{R}^2$

Your score : 0

Discussions (0)

Question 21 : 640653902384

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Consider the function $f(x, y) = x^2 + xy + y^2 - 7x - 8y$.

Your score : 0

Question 22 : 640653902385

View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : SA

If f has a critical point at (a, b), find $a + b$.

Answer (Numeric):

Answer

Accepted Answer : 5

Your score : 0

Discussions (0)

Question 23 : 640653902386

View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : SA

Find the determinant of the Hessian at (a, b).

Answer (Numeric):

Answer

Accepted Answer : 3

Your score : 0

Discussions (0)

Question 24 : 640653902387

View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : MCQ

What is the nature of the critical point?

OPTIONS :

- Local maximum
- Local minimum
- Saddle point

Your score : 0

Discussions (0)

Question 25 : 640653902388

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.
Find three positive numbers x, y, z whose sum is 10 such that x^2yz^2 is maximum.

Your score : 0

Question 26 : 640653902389

View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : SA

Enter the value of x.

Answer (Numeric):

Answer

Accepted Answer : 4

Your score : 0

Discussions (0)

Question 27 : 640653902390

View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : SA

Enter the value of y.

Answer (Numeric):

Answer

Accepted Answer : 2

Your score : 0

Discussions (0)

Question 28 : 640653902391

View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : SA

Enter the value of z.

Answer (Numeric):

Answer

Accepted Answer : 4

Your score : 0

Discussions (0)

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